International Islamic University Chittagong

Morality Development Program (MDP)

Semester End Examination, Spring– 2025

Course Title: Tajweedul Quran I

Course Code: MDP-1201, 2nd Semester

Full Marks : 50 **Time :2.50 Hours**

Answer the following questions

Q.1	**Answer the following questions:** **a. Write the objects of this course.** **b. Write the meaning of** Surah Al-Kauther and Quraesh.	**3** **7**
Q.2	**Discuss kinds of Noon Saakin and Tanween including rules of every kind with example.**	**10**
Q.3	**Identify types of Noon Saakin and Tanween from the underlined words.** مِن رَبِّهِمْ عَذَابٌ عَظِيمٌ مِنْ بَعْدِ هُدًى لِلْمُتَّقِينَ سَمِيعٌ بَصِيرٌ مِنْ دُونِهِ مِنْ حَكِيمٍ **Or** **a.Define Meem saakinah b.Write the kinds of Meem saakinah and its rules.**	**10**
Q.4	**Discuss the terms-Tajweedul Quran, Ghunnah, Idghaam with ghunnah and without ghunnah.** **Or** **Identify types of Meem sakinah from the underline words.** أَفْتَرَىٰ عَلَى ٱللَّهِ كَذِبًا أَم بِهِۦ جِنَّةٌ وَكَذَّبَ ٱلَّذِينَ مِن قَبْلِهِمْ لَهُم مَّا يَشْتَهُونَ	**10**
Q.5	**Write the last 2 Ayats (Verses) from your memory of Surah al-Baqarah with meaning in any language.**	**10**

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Spring–2025

Course Title: Basic Principles of Islam | **Course Code: GEED-1201**
Full Marks: 50 | **Time: 2 Hours 30 Minutes**

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	"`*Ibadah* is a comprehensive phenomenon and it includes all aspects of human life-" justify this statement by explaining some objectives and conditions of *Ibadah*.	**10**	**3**	Create
2	Define *Salah* literally and terminologically. Explain the prerequisites (*Shurut*) and essential elements (*Arkan*) of *Salah* elaborately.	**10**	**3**	Remember & Create
3	Who are the due recipients of *Zakah*? Suggest a proper method of *Zakah* distribution to eradicate poverty from our society.	**10**	**3**	Remember & Create
4	"Fasting is only for me and I will give reward for it"- evaluate this Hadith by explaining the importance of fasting in detail.	**10**	**3**	Create
5	a. "Take the method of *Hajj* from me"- discuss this Hadith by explaining the some methods and their impact on our present life. **Or,** b. "Islam is preached by the character of Muslim; not by terrorism and violence"- analyze this statement by explaining the clear concept of *Jihad* in Islam elaborately.	**10**	**3**	Understand & Create

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M

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc in CSE Semester Final Examination, Spring 2025
Course Code: CSE-1221 **Course Title: Computer programming-II**
Total marks: *50* Time: *2 hours 30 minutes*

Group-A

		Marks	CO	DL
1. a)	What is the role of friend functions in operator overloading? Illustrate with an example.	3	CO2	U
Or,	In C++, can you overload the [] operator ? What are the differences between overloading the subscript operator as a member function vs a friend function? Which one is allowed in C++, and why?			
b)	Find and explain the error in the following C++ code: (code below)	2	CO2	U

```
class Number {
private:
    int x;
public:
    Number(int val) : x(val) {}
    friend int operator = =(Number a, Number b) {
        return a.x = = b.x;
} };
```

		Marks	CO	DL
c)	Design a C++ class **Student** with private members: int id and float cgpa. Overload the following operators: 1. == to check if two students have the **same CGPA** (ignore ID). 2. && to return true if **both students have CGPA ≥ 3.5**.	5	CO2	App
2. a)	Explain the behavior of private and public members where a base class is inherited publicly by a derived class.	2	CO2	U
b)	Create a base class **Vehicle** with a function **displayType().** Derive a class Car from Vehicle with a function **displayBrand().** Then derive a class **ElectricCar** from Car with a function **displayBatteryCapacity().** Show multilevel inheritance and call all display functions from an ElectricCar object.	3	CO2	App
Or,	Create a **base class** named **Shape** with a member function called **area().** Derive two classes: **Rectangle** and **Circle** from **Shape.** Use **function overriding** to implement the **area()** function separately in both **Rectangle** and **Circle,** using the appropriate geometric formulas. The program should **take user input** for the necessary dimensions (length and width for the rectangle, radius for the circle) and **calculate and display the area** for each shape accordingly			
c)	Explain the role of protected access specifier in inheritance. Can a derived class access protected member of its base class's object? Explain with an example or justification.	3	CO2	U
Or,	What are the problems associated with multiple inheritance? How you can overcome this? Explain with an example			
d)	Is there any error in this code? If yes, then correct the code. Display the output.	2	CO2	App

```
class A {                                    class C: public A, public B
   public:                                   {
   void cheers() {
       cout<<"Class A: Hip-hip-hooray";      };
   } };                                      int main()
class B {                                    {
   public:                                      C obc;
   void cheers() {                              obc.cheers();
       cout<<"Class B: Hip-hip-hooray";      }
   } };
```

3. **Group-B**

a)	What is polymorphism in OOP? Distinguish between compile-time binding and runtime binding with their characteristics.	3	CO2	U
b)	Develop a C++ program that models an online shopping system using inheritance and polymorphism. Create a base class Product, and derive classes like Electronics, Clothing, and Grocery. Use virtual functions to calculate the price after applying different discounts.	3	CO3	Ap
Or,	Create a base class Employee with a virtual function **displaySalary().** Derive two classes: **PermanentEmployee** and **ContractEmployee.** Override the **displaySalary()** function in both derived classes to compute and display salary according to their respective payment models.			

c) Write the output of the following program: 2 CO2 U

```
class Fruit {
public:
    virtual void taste() {
        cout <<"Fruit taste function \n";
    }
    virtual void eat() {
        cout <<"Fruit eat function \n";
        this->taste();
    }
};

class Apple : public Fruit {
public:
    void taste() override {
        cout <<"Apple taste function \n";
    }
    void eat() override {
        cout <<"Apple eat function \n";
        this->taste();
    }
};

int main() {
    Apple *f = new Fruit;
    f->eat();
}
```

d) Explain the concept of pure virtual function. Write a program with an abstract class containing at least one pure virtual function. 2 CO2 U

4.

a) Suppose you have a program that finds the minimum value between two variables of different data types: integers, floating-point numbers, and characters. The original program uses separate functions for each data type (e.g., Min(int, int), Min(double, double), Min(char, char)). 4 CO3 App

Your task is to re-factor the code using a generic function called Min. The Min function should be capable of finding and returning the minimum of two values of any data type.

Or, You are given a program that searches for a specific element in an array. The original implementation includes separate search functions for different data types — one for integers, one for floats, and another for characters.

Your task is to refactor the program using a generic function named **searchElement**. The **searchElement** function should take an array, its size, and a target value, and return whether the value is found in the array regardless of the data type.

b) Write a C++ program that validates a person's age for voter registration. The program should: 3 CO3 App

1. Accept age as input from the user.
2. If the age is less than 18, throw an exception with a suitable error message.
3. Use a try-catch block to handle the exception and display the message.
4. If the age is valid (18 or above), display "Eligible to vote".

c) Write a C++ program to perform the following operations on a vector. Start with the vector initialized as {11, 13, 17, 19, 23}. Then: 3 CO3 App

1. Insert the number 9 at the **start** of the vector.
2. Append the number 29 at the **end** of the vector.
3. Delete the element at **index 3**.
4. Replace the element at **index 2** with 15.
5. Sort the vector in **descending order**.
6. Finally, display all the elements of the vector.

5.

a) Write a C++ code using manipulator to provide the following output specification for printing float value. 3 CO3 Ap

- 14 column widths
- Right justified
- Three precisions
- Filling unused place with sign
- Trailing zeros shown are display + sign as first position.

Or, Using I/O manipulators, write a program to display a floating-point number with:

- Width: 10
- Precision: 3
- Left-justified and filled with *.

b) What is the difference between formatted and unformatted input/output in C++? Write a program to illustrate both types of I/O operations. 3 CO2

c) In the Squid Game series, the **FrontMan** needs to store participant information. Create a C++ program that will help the **FrontMan** store participant details (name, ID, location) in a file called "SG_S2.txt", then retrieve and display this information on the console. 4 CO3

Or, Write a program to read employee details **(name, ID, salary)** from the user and write them to a file named **employees.txt**. Then read and display the file contents.

International Islamic University Chittagong
Department of Computer Science and Engineering

Final Examination, Spring - 2025
Course Code: **EEE-1221**
Time: 2 hour 30 minutes

Program: B.Sc. Engineering in CSE
Course Title: **Electronics**
Full Marks: 50

[Answer all the questions from the following. Figures in the right margin indicate full marks]
Course Outcomes (COs) and Blooms Levels are mentioned in additional columns.

Part-A

1)	a)	Define JFET. Explain the construction and working principle of n-channel JFET with a neat diagram.	CO2	U	5
		OR			
		Draw and explain the input and output characteristics of EMOSFET.			
	b)	A JFET has a drain current of 5mA. If I_{DSS}=10mA and $V_{GS(off)}$ = -6V, Find the value of i) $V_{GS}$and ii) V_P	CO3	A	5
2)	a)	What is a multivibrator in electronics? With neat diagram explain the operational principles of Monostable multivibrator.	CO2	U	5
		OR			
		What is a switching circuit? Explain the switching action of a transistor with the help of output characteristics.			
	b)	Fig-2(b) shows the transistor switching circuit. Given R_B= 2.7kΩ V_{BB}=3V, V_{BE}=0.7V and V_{knee}=0.7V i) Calculate the minimum value of β for saturation ii) If V_{BB} is changed to 1V and transistor has minimum β=70, will the transistor be saturated.	CO3	A	5

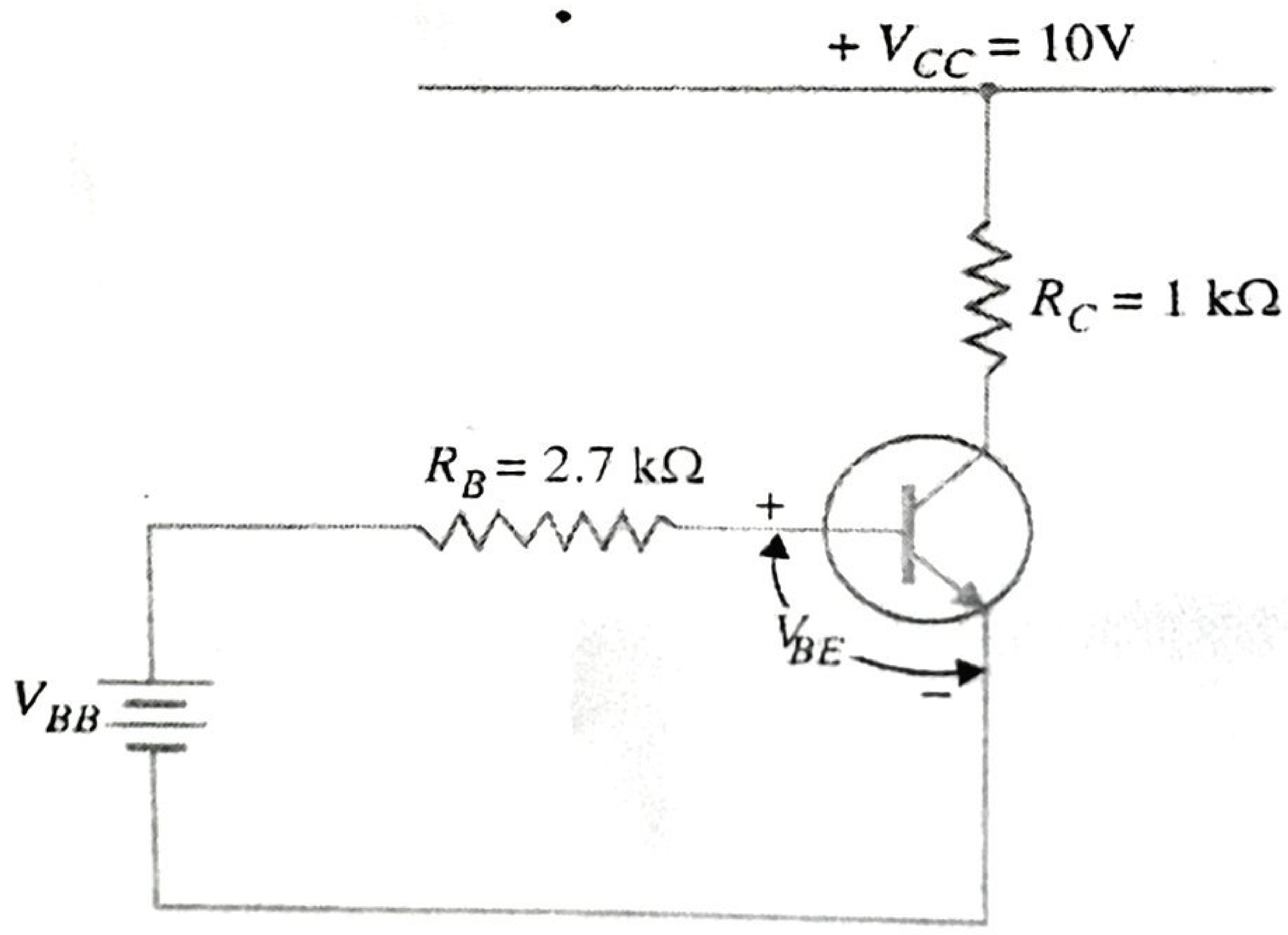
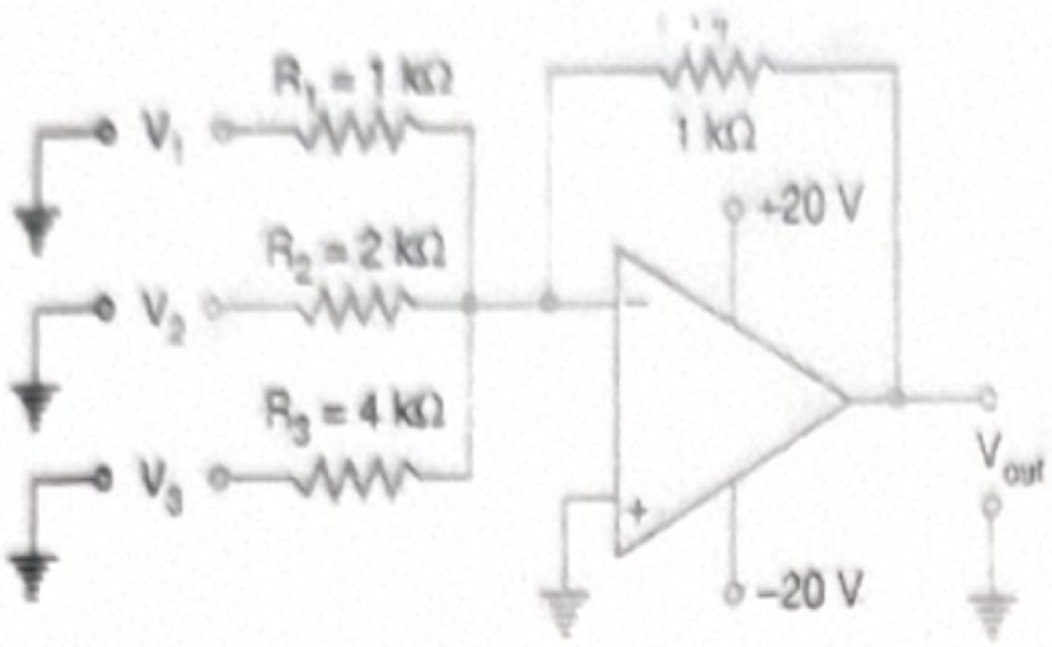


Fig. 2(b)

Part-B

3) a) What are the ideal characteristics of Op-Amp? Derive the expression for the summing amplifier operationusing inverting amplifier?" CO2 U 5

b) Determine the output of the circuit in Fig. 3(b) for different input voltages in Table 3(b). CO3 A 5



V_1(V)	V_2(V)	V_3(V)
+10	0	+10
0	+10	+10
+10	+10	+10

Fig. 3(b) Table. 3(b)

4) a) What is negative feedback? Derive the gain of negative voltage feedback amplifier. CO2 U 5

OR

What is an electronic oscillator? Describe the construction and working principle of a Hartley oscillator.

b) When negative voltage feedback is applied to an amplifier of gain 100 the overall gain falls to 50. CO3 A 5

i) Calculate the fraction of the output voltage feedback.

ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

5) a) Explain the working of a Comparator using Op-Amp with a proper circuit diagram CO2 U 5

b) Describe the working of a Precision Rectifier using Op-Amp.

OR

Analyze the output voltage in the context of Fig. 5(b), where an op-amp integrator is connected to a square wave input. CO3 A 5

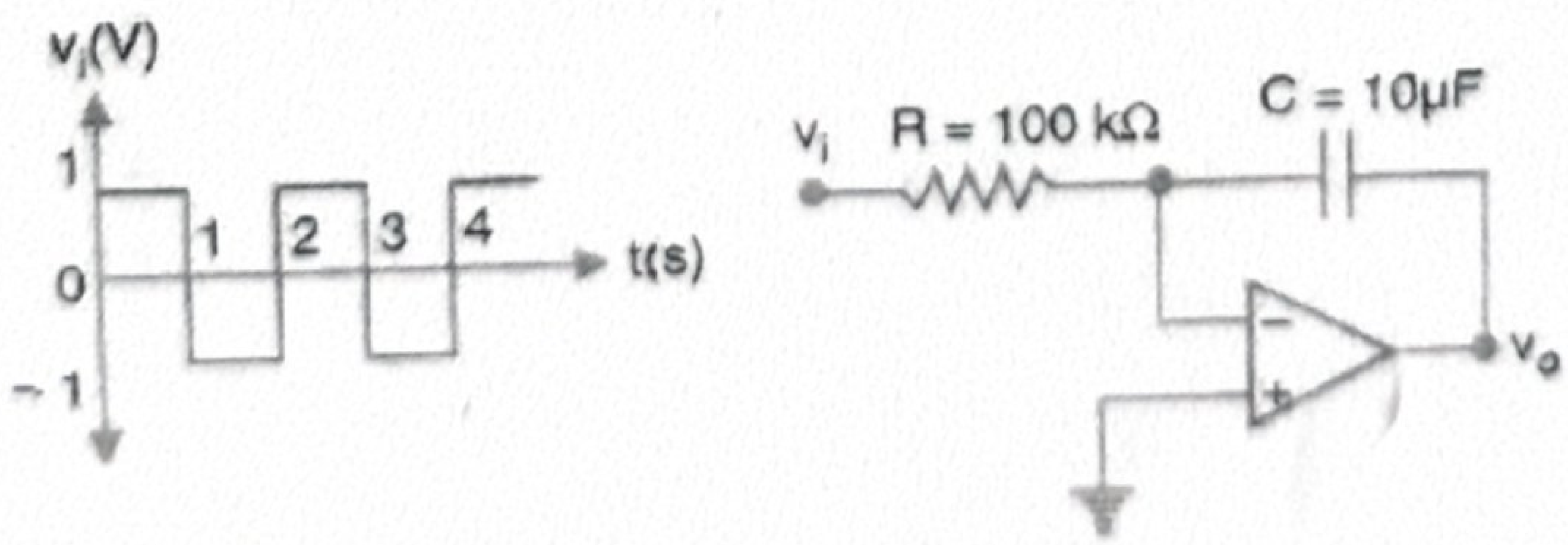


Fig. 5(b)

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester Final Examination

Program: B. Sc. in CSE
Course Code: MATH-1207
Time: 2:30 hours

Semester: Spring-2025
Course Title: Mathematics-II
Total Marks: 50

(i)	Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii)	Please answer the several parts of a question sequentially.
(iii)	Separate answer script must be used for separate group.
(iv)	Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO2:	Solve differential equations using various methods.
CLO3:	Formulate the mathematical model and interpret the results by analyzing the real-world problems related to Growth and Decay Problems, Temperature Problems, Falling Body Problems, Dilution Problems, Electrical Circuits problems etc. through a set of differential equations.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group – A

		Question	Marks	CLO	DL
1.	a)	Define differential equation with example. Form a differential equation whose solution is given by $y = ax + bx^2$. Hence mention the order and degree of that equation.	1 3 1	CLO2	U
	b)	Solve the following equations (i) $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ (ii) $(6x - 4y + 1)\frac{dy}{dx} = (3x - 2y + 1)$	2 3	CLO2	U
	Or)	Solve the following equations (i) $\frac{dy}{dx} + \frac{2x+3y+1}{3x+4y-1} = 0$ (ii) $(x - y)^2 \frac{dy}{dx} = a^2$			
2.	a)	Define Bernoulli's differential equation. Solve the Bernoulli's differential equation, $\frac{dy}{dx} + xy = x^3 y^3$	5	CLO2	U
	b)	Test whether the differential equation, $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$ is exact or not hence solve it.	5	CLO2	U
	Or)	Solve the linear differential equation with constant coefficients $\frac{d^3r}{dt^3} + 3\frac{d^2r}{dt^2} + 3\frac{dr}{dt} + r = e^{-t} + 1 + \text{Sin } 2t$			

Group – B

			Marks	CLO	DL
3.	a)	Define Bessel's equation. Using Bessel's function prove that, $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$	5	CLO2	U
	b)	Solve the linear differential equation by the **method of Variation of parameters** $(D^2 - 1)y = 2/(1 + e^x)$			
Or)	a)	Define Legendre's equation. Using Rodrigue's formula. Show that $\int_{-1}^{1} [P_n(x)]\, dx = \begin{cases} 0, \text{when } n \neq 0 \\ 2, \text{when } n = 0 \end{cases}$	5	CLO2	U
	b)	Solve the linear differential equation by the **method of Undetermined Coefficients,** $(D^2 - 4D - 5)y = x^2 + 2e^{3x}$			
4.	a)	Define partial differential equation. Solve the linear partial differential equations by **Lagrange's method,** $yzp + xzq = xy$	5	CLO2	U
	b)	Solve the non-linear partial differential equations by **Charpit's method,** $px + qy = pq$	5	CLO2	U
5.	a)	36th July 2024, an activist of the 'Anti-Discrimination Students Movement' was seriously injured at Jatra Bari, Dhaka. Other activists brought him to Dhaka Medical College Hospital at 11:00 pm, and the doctor discovered that he was dead. The doctor took the temperature of the body at 11:30 pm, which was 94.6°F. He again took the temperature after one hour, which showed 93.4°F, and noticed that the temperature of the morgue of DMCH was 70°F. If the normal temperature of the human body is 98.4°F, then form a differential equation, and by solving it, estimate the time of death.	5	CLO3	Ap
	b)	A generator having emf 100V is connected in series with a 10Ω resistor and an inductor of 2H. If the switch k is closed at time $t = 0$, Obtain a differential equation for the current and determine the current at time t.	5	CLO3	Ap

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Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring-2025
Course Code: PHY-1201 **Course Title: Physics-II**
Total marks: ***50*** Time: *2* hours *30* minutes

[Answer ***all*** the questions. Figures in the right hand margin indicate full marks.
Separate answer script must be used for Group A and Group B]

Group-A

1. a) Define: i) Lattice ii) Crystal — CO1 R 2

   b) Define packing fraction. Derive the expression for the packing fraction of a face-centered cubic (FCC) lattice and show that its value is approximately **0.74**. Explain the physical significance of this value. — CO2 U 5

   Or

   Define coordination number. Explain coordination number for NaCl structure.

   c) Lead is face-centered cubic with an atomic radius of $r = 1.564$ A.U. Find the spacing of (212) planes. — CO3 A 3

2. a) Define Miller indices in crystallography? — CO1 U 2

   b) Show that in a crystal of cubic structure, the distance between the planes with Miller indices h, k, l is equal to $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$, where a is the lattice parameter. — CO2 R 5

   Or

   Distinguish between metals, semiconductors, and insulators based on their energy band structure.

   c) Draw the following planes: (010), (101), and (222). — CO3 A 3

Group-B

3. a) State the postulates of the special theory of relativity. — CO1 R 2

   b) Demonstrate that a moving clock appears to run slower compared to a stationary clock. — CO2 U 5

   c) The length of a spaceship is measured to be exactly half its actual length. Calculate — CO3 A 3
      (i) the speed of the spaceship
      (ii) the time dilation corresponding to one second on the spaceship.

4. a) What is an X-rays? Write down the properties of X-rays. — CO2 U 3

   b) Calculate the following properties of ~~alpha particle~~ $(^{235}_{92}U)$: — CO3 A 7
   Uremium
      i. Nuclear Mass
      ii. Nuclear Size
      iii. Nuclear Density
      iv. Nuclear Charge
      v. Nuclear Mass defect
      vi. Nuclear Binding Energy
      vii. Nuclear Binding Energy per nucleon.

5. a) State the laws of radioactive disintegration. Show that, the number of radioactive atoms decrease exponentially with time. — CO2 U 7

   or

   Derive the expression that relates the mass of a moving object to its rest mass according to the theory of special relativity.

   b) A fossil is found to have 12.5% of its original Carbon-14 content remaining. The half-life of Carbon-14 is 5730 years. How old is the fossil? — CO3 A 3

   or

   A beam of X-ray is scattered by a free electron at 35^0 from the direction of incidence. The scattered X-ray has a wavelength of 0.044 Å. What is the wave-length of the incidence X-ray?

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International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Spring-2025

Course Code: CSE-1223 **Course Title: Discrete Mathematics**

Total marks: *50* Time: *2* hours *30* minutes

[Answer *all* the questions; in some questions, there are options; you will solve any one of them.
Figures in the right-hand margin indicate full marks. Separate answer script must be used for Group-A and Group-B]

Group-A

1a) What is a *composite number*? Show that if **n** is a composite integer, then **n** has a *prime divisor* less than or equal to √**n**. 3 CLO2 U

1b) What do you mean by *linear congruence*? Solve the following linear congruence equation: 3 CLO2 A

`48x ≡ 28 (mod 355)`

Or, Express `gcd(119,544)` as a linear combination using the *Euclidean algorithm.*

1c) Find the value of **X** using the *Chinese Remainder Theorem* 4 CLO2 A

`X ≡ 1(mod4)`

`X ≡ 2(mod3)`

`X ≡ 3(mod5)`

Or, Explain the *encryption* and *decryption* functions of the *Caesar Cipher*. Encrypt the word **CAT** using the Caesar Cipher with key = **3**. (Use `A=0, B=1, ..., Z=25`)

2a) State the 1st and 2nd principles of *mathematical induction*. Why is mathematical induction valid? 3 CLO1 U

2b) Use *mathematical induction* to prove that: $4 + 4{\cdot}6 + 4{\cdot}6^2 + \dots + 4{\cdot}6^n = \frac{4(6^{n+1}-1)}{5}$, whenever n is a nonnegative integer. 4 CLO1 Ap

Or, Prove that √3 is irrational by giving a *proof by contradiction.*

2c) Let the function `g(n)` be defined *recursively* as: `g(0) = 2, g(1) = 3, g(n+1) = 2g(n) + g(n−1),` for `n ≥ 1.` Find: `g(2) and g(4).` 3 CLO1 A

Group-B

3a) In how many ways can 7 females and 10 males be arranged in a row such that: 3 CLO2 A

i) All females stand together?

ii) All males stand together?

3b) State *Inclusion-Exclusion Principle*. Each user on a computer system has a password, which is 6 to 8 characters long, using uppercase letters and digits. It must contain at least one digit. How many possible passwords are there? 4 CLO2 A

Or, State Inclusion-Exclusion Principle. A security system generates **10**-digit access codes made up of only **0**s and **1**s (bit strings). How many different access codes are possible that either *start with the sequence 101 or end with a 1?*

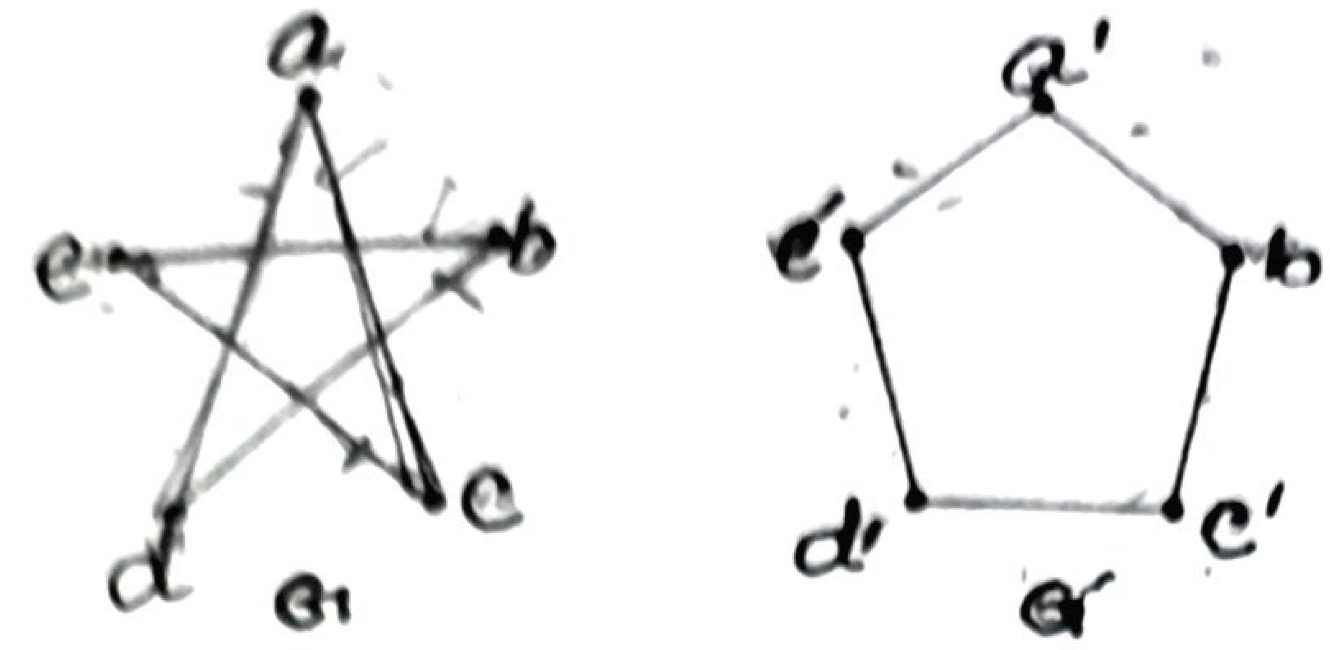
3c) State *the generalized pigeonhole principle*. What is the minimum number of investors required to ensure at least **25** are allocated to one district, if there are **7** districts? 3 CLO2 U

4a) Write short notes on: 2 CLO3 U

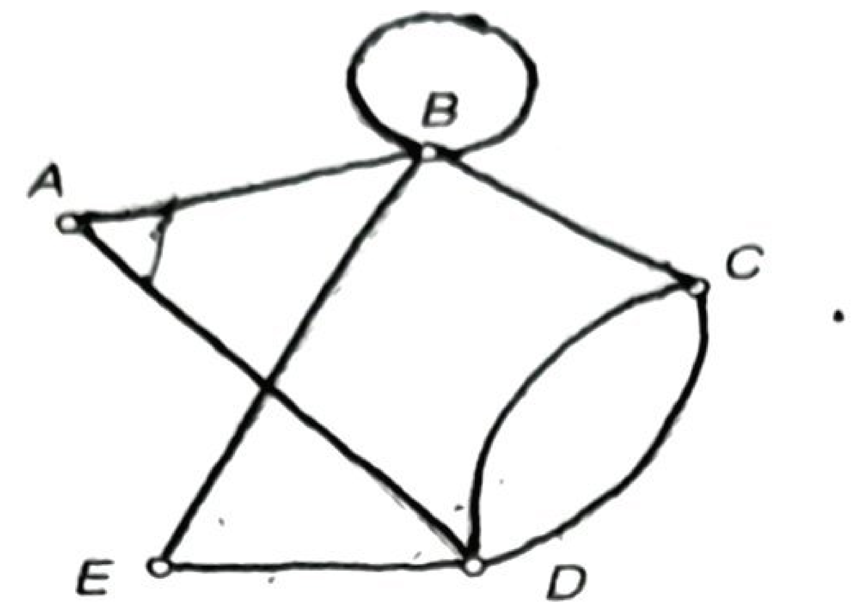
i) Complete Graph ii) Bipartite Graph

Or, Draw two graphs: K_5, $K_{3,4}$

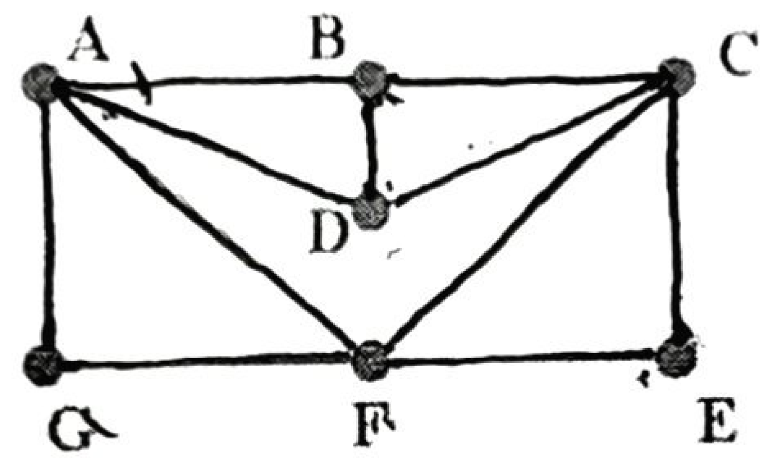
4b) What do you mean by *isomorphism of graphs*? Determine whether the following two graphs are isomorphic or not. 4 CLO3 Ap



4c) What do you mean by *Euler circuit* and *Euler* path? Determine whether the following graph has a *Euler circuit* or *Euler* path. Construct such a circuit or path if exists. **4** **CLO3**

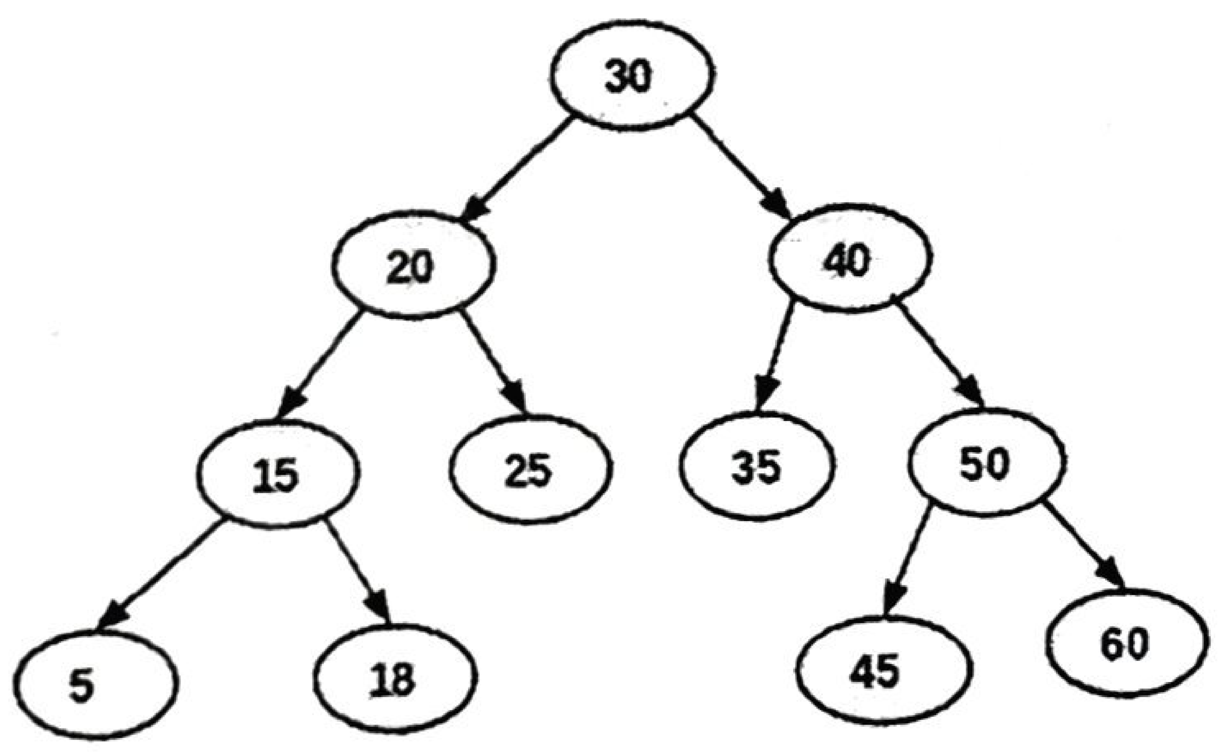


Or, What do you mean by *Hamiltonian circuit* and *Hamiltonian path*? Determine whether the following graph has a *Hamiltonian circuit* or *Hamiltonian path*. Construct such a circuit or path if exists.



5a) Define *tree*. Write down the differences between *graph* and *tree*. **2** **CLO3**

5b) Find *inorder*, *preorder*, and *postorder* traversals of the following tree. **3** **CLO3**



5c) Define *spanning tree* and *minimum spanning tree*. Construct a minimum spanning tree of the following graph. **5** **CLO3**